SPPA-MVFit Submission Supplement: Protocol, Sealed Results, and Claim Boundaries

Purpose

This short supplement is the formal companion for the SPPA-MVFit paper. The longer historical technical diary is **not** submitted. This document records the sealed confirmatory protocol path, key hashes, and claim boundaries. Primary scientific tables appear in the main paper.

Primary endpoint (held-out)

- Estimand: clean 64-cubed voxel IoU of `sppa_mvfit` minus `generic_mvfit`.
- Units: 240 source actors (12 family$\times$stratum cells, equal weight).
- Decision: stratified bootstrap 95% CI lower bound $> +0.030$.
- Result: mean $\Delta = 0.190$, CI $[0.181, 0.199]$, **H1 PASS**.
- Strata: CSG-ID 0.209, implicit-OOD 0.172 (both positive).
- Provenance: `synthetic_geometry` only.

Reproduction commands

Run from the git repository root that contains the `paper_semantic_proxy_3d` junction (or paper directory). Do not re-optimize the sealed held-out methods after inspection.

Table 1: Minimal checks for the submitted primary claim.

Command	Purpose	Expected
`python-mpytestpaper_semantic_proxy_3d/reproducibility/sppa_mvfit/tests/test_contract.py`	Shared builder, budget, no cross-imports, split balance.	7 passed
`pythonpaper_semantic_proxy_3d/reproducibility/sppa_mvfit/benchmark/verify_package.py--development`	Development package integrity.	PASS
`pythonpaper_semantic_proxy_3d/reproducibility/sppa_mvfit/benchmark/export_paper_tables.py`	Regenerate main-paper tables from sealed `raw_metrics.csv`.	writes `.tex`
`pythonpaper_semantic_proxy_3d/tools/reproduce_sppa_mvfit_paper.py--strict`	Hash/gate checks for the submission snapshot.	0 blockers

Held-out generation/evaluation commands exist in `reproducibility/sppa_mvfit/README.md` and require the protocol PASS record, pretest freeze, and NIST-bound seed manifest. Re-running method optimization for a better score is forbidden by protocol.

Selected sealed hashes

- Pretest freeze SHA-256: `8E2ADBF32F299B24CD2A5AB87C74D142E707696F79D18DF0C3332209C3B46CA3`
- Protocol PASS SHA-256: `2348946BDDB04B8E5CA7D2C845C5F5C45F1AE06F8907E99218ED5E9A379FA74F`
- Sealed predictions binary SHA-256: `F870C57D9CC6FF4868EFB25FD2926FA7D19858EAF8CD0E9781F38990D7D145FD`
- Raw metrics SHA-256: `57A82D234F55013D76BEF2E36CF2B3F7C5617DD4FA6EF811C2A8447A04C0AD63`

Claim boundaries

Supported: family-conditioned occupancy gain under synthetic calibrated silhouettes; shared generator; optional runtime contract evidence.

Not supported: measured flight, real-UAV 3D GT accuracy, operator benefit, universal open-set geometry, or neural image-to-3D SOTA ranking.

Real-image figures in the main paper and in this supplement remain qualitative probes without 3D ground truth and use declared replay assumptions where telemetry is shown.

Supplementary figures

The two audit artifacts below are referenced from the main paper as Supplementary Figures S1 and S2. They duplicate no main-paper figure: S1 provides the zoomed detector-evidence detail that the compressed comparison grid cannot show, and S2 provides the multi-view renders behind the visual part evidence audit table.

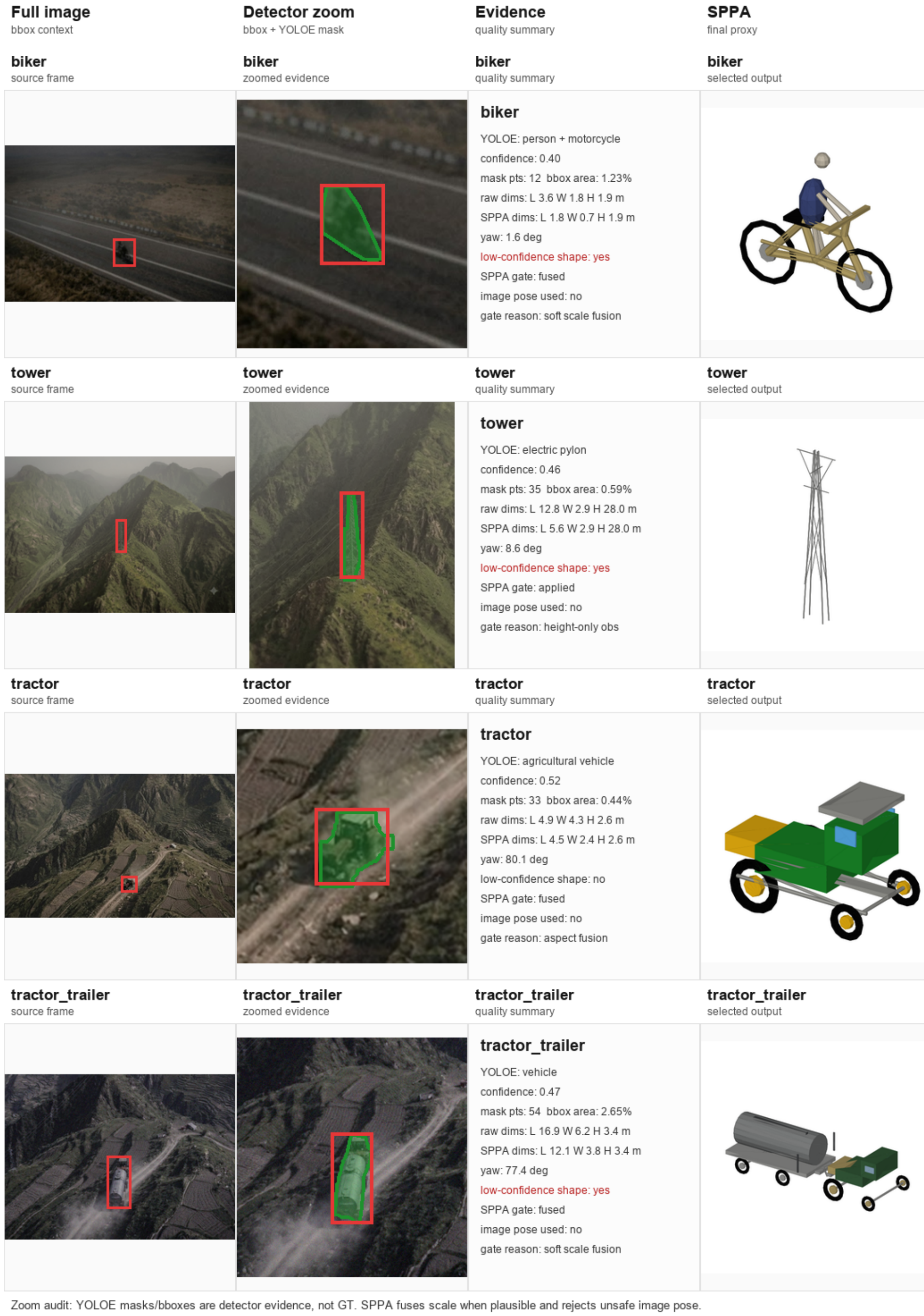

Figure S1: Zoomed real-input audit for the four user-supplied probes. Red boxes and green masks are YOLOE detector evidence. The evidence panel reports confidence, mask size, bbox area, inferred replay dimensions, yaw, and the SPPA low-confidence shape flag. The zoom panel shows the final selected proxy; the separate input-mode ablation reports text-only, detector+metric, and detector+metric+visual variants. The evidence panel reports raw detector dimensions and the final SPPA dimensions, showing that the tractor's nearly square footprint is not copied but corrected through the vehicle aspect prior, while low-confidence image pose is not used. SPPA recipes include explicit connector primitives for semantically attached parts, such as bicycle frame tubes, tower braces, tractor axles, and tractor-trailer hitch/axle connectors. The full frames in the first column are also the real-image replay evidence: red boxes are the selected YOLOE or composite detector bboxes and green polygons are the native YOLOE/Ultralytics mask contours used by the observation bridge, which remain detector evidence rather than ground-truth masks.

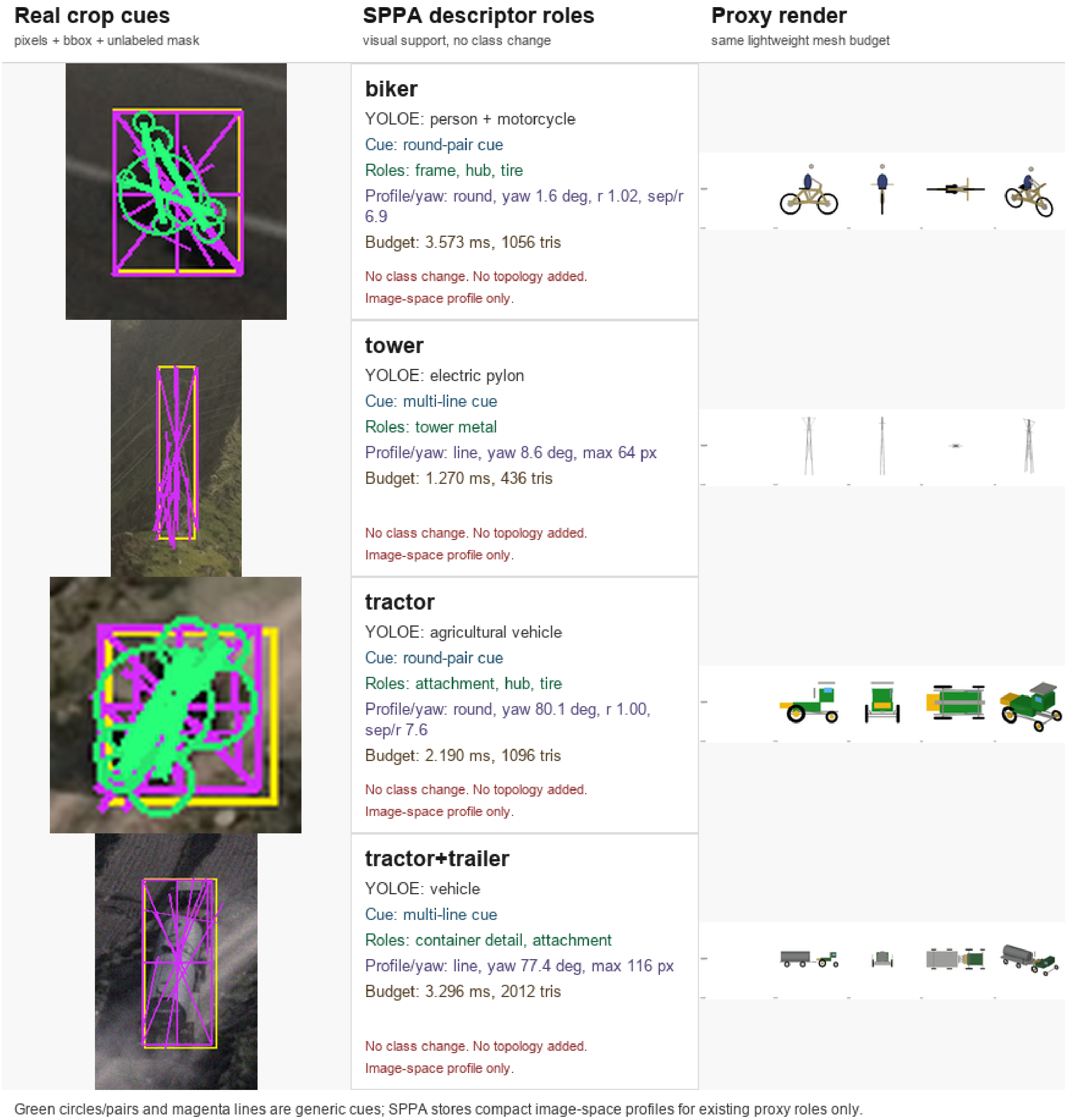

Figure S2: Visual part evidence bridge on the four real probes. Green circles/pairs and magenta lines are generic image-space cues from pixels, detector boxes, and unlabeled mask geometry. SPPA maps those cues only onto descriptor roles already present in the selected proxy and records a compact image-space geometry profile; when the visual–metric gate is non-divergent, the selected yaw is the declared-replay axial footprint yaw. The semantic class and mesh topology are unchanged.